Appendix: PolSAM Supplementary Material

I. PhySAR-Seg-1 Dataset Clustering

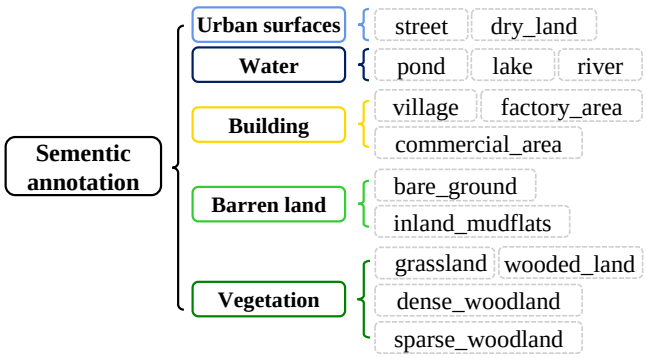


Fig. 1. Semantic label clustering of data PhySAR-Seg-1.

Initially, the dataset contains 14 land cover types, which we merge into five categories: urban surfaces, water, buildings, barren land, and vegetation, as shown in Fig. 1.

II. Cross-band Consistency Analysis of MVD

To quantify the cross-band consistency of MVD, the four MVD products of the PhySAR-Seg-1 dataset, which cover the same geographic region, are co-registered to the coarsest (L-band) grid via nearest-neighbor resampling, cropped to the common valid region, and aggregated by majority voting within non-overlapping 8×8 blocks to suppress speckle noise. The registration accuracy is verified by the agreement of the four annotation maps, which exceeds 96.8% for all band pairs. On this basis, we compute the pairwise agreement (Agr.) and Cohen's kappa (κ), the standard correlation measure for categorical maps, at both the 8-class level (excluding the undefined edge category) and the primary scattering mechanism level (odd-bounce, double-bounce, volume, and mixed).

In addition, the alignment between each band's MVD and the semantic annotation is measured by purity, i.e., the fraction of pixels covered by the majority semantic class of each cluster, and normalized mutual information (NMI, 0 for independent and 1 for identical partitions). Since scattering mechanisms and land covers form a many-to-many correspondence, the absolute values of both metrics are inherently bounded and serve for relative comparison across bands.

As shown in Table I, the mechanism-level agreement is consistently higher than the 8-class agreement, and both are substantial among the shorter-wavelength bands (S–C: 69.84%, C–X: 67.39%, S–X: 57.04%, with κ up to 0.441) while dropping for all pairs involving the L band (32.19–43.77%). Among the mechanisms, odd-bounce regions such as water bodies and smooth surfaces are the most stable across all pairs (54.65–92.60%), and double-bounce regions agree well among the S, C, and X bands (55.99–76.74%) but rarely for the L-band pairs. The qualitative comparison in Fig. 2 explains this pattern: the spatial structures and boundaries of ground objects are highly consistent across the four bands, whereas the mechanism assigned to the same object shifts systematically with wavelength—croplands appear volume-dominated at the L band owing to canopy penetration but surface-dominated at shorter wavelengths, and the visibility of building double-bounce responses also varies with wavelength.

TABLE I
Pairwise cross-band consistency of MVD on PhySAR-Seg-1. Agr.: pixel-level agreement; κ: Cohen's kappa.

Band pair	MVD classes (8)		Scattering mechanisms (4)	
	Agr. (%)	κ	Agr. (%)	κ
L–S	22.21	0.027	32.19	0.048
L–C	29.64	0.085	40.97	0.116
L–X	35.35	0.204	43.77	0.186
S–C	65.46	0.391	69.84	0.375
S–X	44.16	0.125	57.04	0.240
C–X	49.37	0.280	67.39	0.441

TABLE II
Alignment between each band's MVD and the semantic annotation on PhySAR-Seg-1.

	BandL	BandS	BandC	BandX
Purity (%)	55.16	44.83	47.47	49.41
NMI	0.153	0.094	0.176	0.156

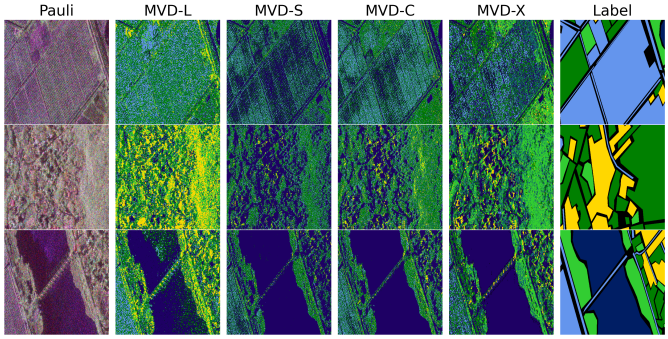


Fig. 2. Qualitative cross-band comparison on three representative regions (rows: cropland, built-up area, and water). Columns: Pauli pseudo-colored image, MVD of the four bands, and the semantic label. Spatial structures are highly consistent across bands, while the mechanism assignments vary systematically with wavelength.

Despite these wavelength-dependent assignments, Table II shows that all four bands yield a comparable alignment with the semantic annotation (purity 44.83–55.16%, NMI 0.094–0.176), indicating that the generation pipeline produces semantically meaningful MVD products for every band; the selection of the X band for the released dataset is discussed in Section III-B of the main text. Overall, the cross-band

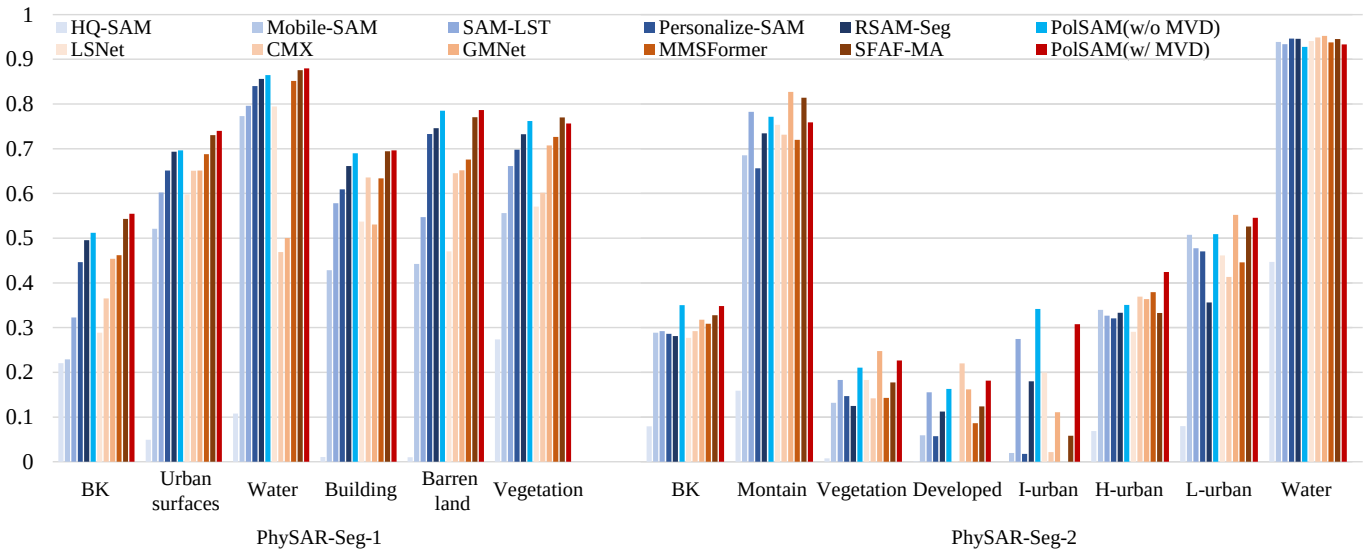


Fig. 3. Comparison of IoU metrics for each category across different models on two datasets. The left side corresponds to the PhySAR-Seg-1 dataset, and the right side corresponds to the PhySAR-Seg-2 dataset.

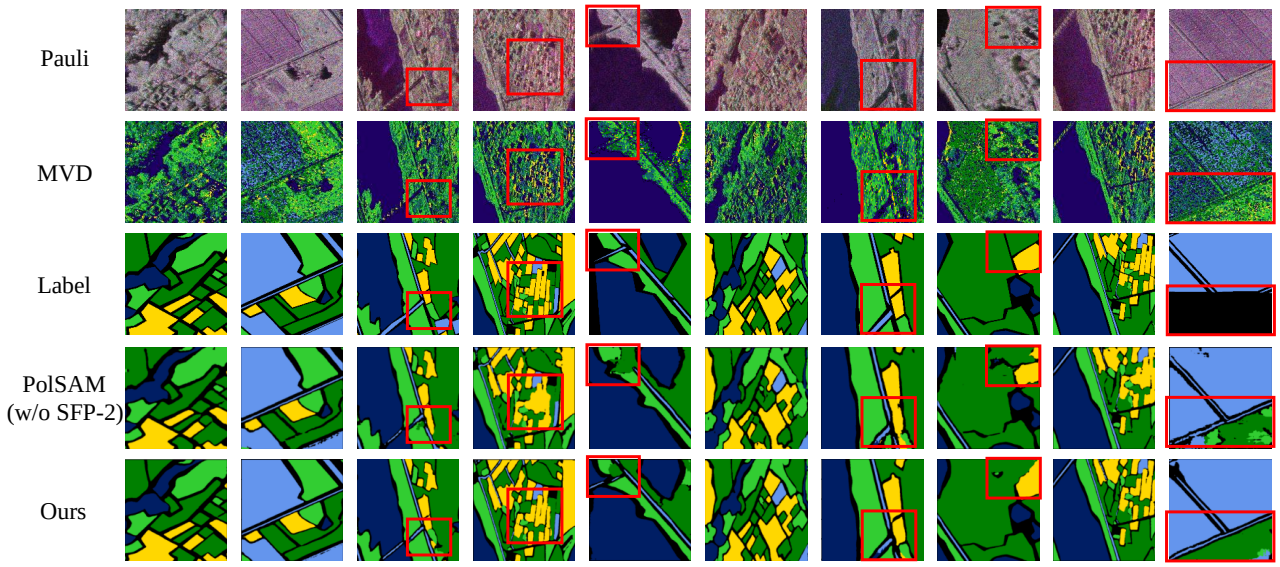


Fig. 4. Qualitative segmentation results on the PhySAR-Seg-1 dataset: each row displays a randomly selected image from the test set, while each column compares the outputs of different models.

differences reflect systematic wavelength effects inherent to electromagnetic scattering rather than instability of the MVD generation pipeline, which further justifies the semantic re-clustering step and the single-band MVD product adopted in our pipeline.

III. Experiments

A. Per-category visualization

The metrics for each category of the two datasets are visualized in Fig. 3, providing a clear comparison of the results. The blue bars represent our PolSAM(w/o MVD) method, corresponding to the blue-themed SAM-based approaches, while the red bars indicate our PolSAM(w/ MVD) method, aligned with the red-themed multimodal methods. Our methods demonstrate three key strengths. First, both PolSAM (w/ MVD) and PolSAM (w/o MVD) consistently outperform other methods across most categories. Second, PolSAM (w/ MVD) generally delivers better overall performance than PolSAM (w/o MVD), highlighting the advantage of incorporating MVD. Finally, for the PhySAR-Seg-1 dataset, our method achieves the best results across nearly every category. For the PhySAR-Seg-2 dataset, particularly in the urban class where double-bounce scattering is dominant but the class proportion is small, our method demonstrates significant improvements compared to other approaches. This indicates that, even in challenging scenarios, effectively leveraging MVD information can substantially enhance segmentation accuracy.

B. Ablation Studies on SFP-2

To validate the effectiveness of the progressive design of the SFP module, we conducted an additional ablation study on SFP-2, referred to as M5. The visualization results on the PhySAR-

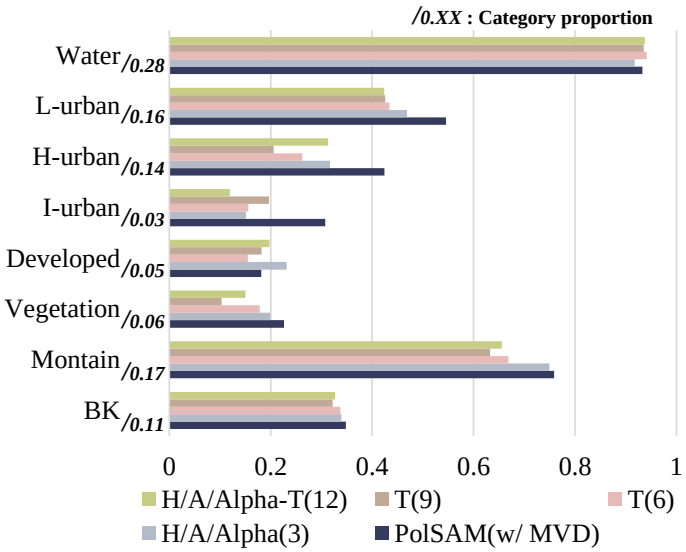


Fig. 5. Impact of physical information input forms on IoU per category in the PhySAR-Seg-2 dataset.

Seg-1 dataset are presented in Fig. 4. Our method, PolSAM (w/ MVD), which integrates SFP-2, significantly outperforms the version without SFP-2, especially in the details highlighted by the red box. It is important to note that the labels in the last column of images are incomplete; however, our results still accurately predict the same semantic categories as the original images. This indicates that the progressive design effectively enhances the interaction between multi-input features and high-level semantic features, enriching the semantic information embedded in the fusion prompt.

C. Discussions on Effectiveness of MVD.

To validate the effectiveness of MVD, we conducted experiments comparing different polarimetric decomposition features as substitutes for MVD in the PolSAM.

The H/A/Alpha decomposition method is used to analyse scattering mechanisms in PolSAR data. The entropy (H) and alpha angle (α) are calculated using the eigenvalues and eigenvectors of the coherency matrix. This allows for the characterisation of surface, double-bounce, and volume. The entropy (H) is calculated using the formula [$H = -\sum_{i=1}^{3} p_i \log_3(p_i)$], where p_i represents the probabilities associated with the eigenvalues. The alpha angle (α) is calculated using the formula [$\alpha = \sum_{i=1}^{3} p_i \alpha_i$], where α_i represents the corresponding scattering mechanism angles. This technique offers the benefit of offering a comprehensive explanation of scattering mechanisms, but it may be restricted by its susceptibility to noise and the assumption of simplistic scattering models, which may not accurately represent more complex ground features.

The polarization coherence matrix T is derived from the Pauli scattering vector under the reciprocity condition and represents the fully polarimetric information in PolSAR imaging. The matrix is given by:

$$T = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} = \mathbf{k}_{\mathrm{P}} \mathbf{k}_{\mathrm{P}}^{H},$$

where $\mathbf{k}_{\mathrm{P}} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} + S_{\mathrm{VV}} & S_{\mathrm{HH}} - S_{\mathrm{VV}} & 2S_{\mathrm{HV}} \end{bmatrix}^T$ is the Pauli scattering vector and $\mathbf{k}_{\mathrm{P}}^{H}$ is its conjugate transpose. The elements T_{ij} are the entries (i, j) of the matrix T, and S_{HV} denotes the rescattered return of the horizontal transmitting and vertical receiving polarizations. The polarization coherence matrix T is a symmetric matrix, meaning its upper triangular elements can fully represent its polarimetric features. As a result, the decomposition of T yields nine components: T_{11}, T_{22}, and T_{33} (the diagonal elements representing the power in each polarization channel), Re$[T_{12}]$, Im$[T_{12}]$ (the real and imaginary parts of the cross-polarization components), Re$[T_{13}]$, Im$[T_{13}]$ (additional cross-polarization components), and Re$[T_{23}]$, Im$[T_{23}]$ (remaining cross-polarization components).

The real parts of these upper triangular elements form the six-channel T(6) dataset, the full set of real and imaginary components together constitute the nine-channel T(9) dataset. Additionally, the H/A/Alpha_T(12) approach combines the three channels from the H/A/Alpha decomposition with the nine-channel data from the T decomposition, resulting in a comprehensive twelve-channel dataset.

To further evaluate the effectiveness of our method, we conducted a detailed analysis of IoU metrics for each class in the PhySAR-Seg-2 dataset. The results, visualized in Fig. 5, show that our method using MVD achieves the best results across most classes, particularly for the urban class, which is characterized by double-bounce scattering and has the lowest overall proportion in the dataset.